CHAPTER 6

Data Link Layer

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6.1 Link Layer: Introduction and Services

- **Data Link Layer Services:** Framing, Link access

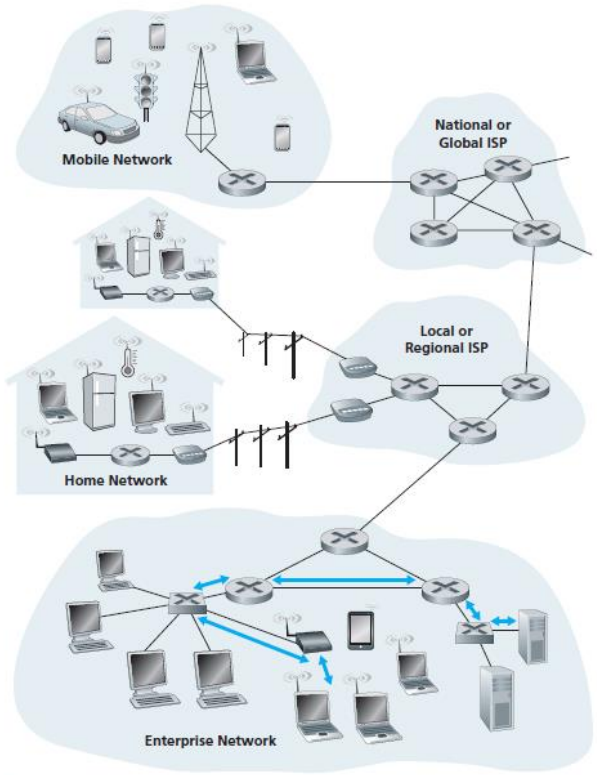


Figure 6.1 ♦ Six link-layer hops between wireless host and server

6.3 Multiple Access Protocols

- **Channel Partitioning:** FDMA, TDMA
- **Random Access:** CSMA/CD
- **Taking Turns:** Token ring

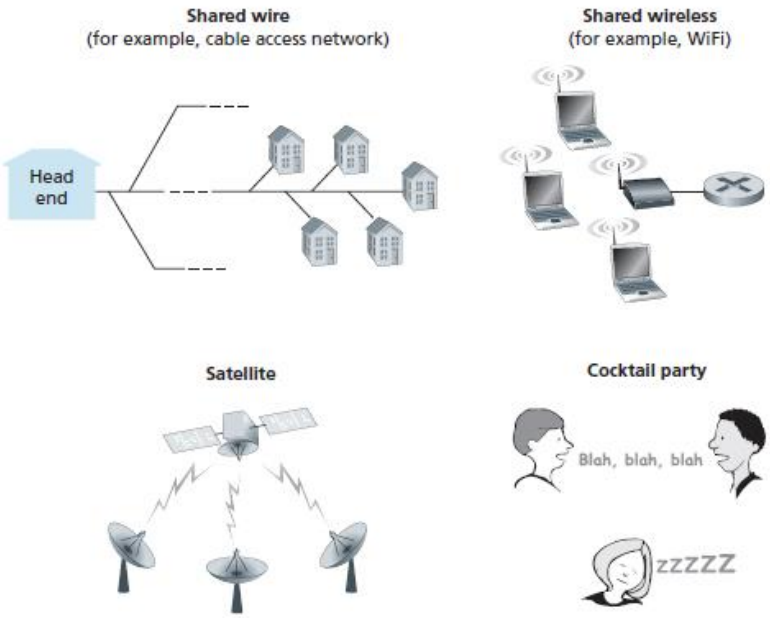


Figure 6.8 ♦ Various multiple access channels

6.3.2 Random Access Protocols (CSMA/CD)

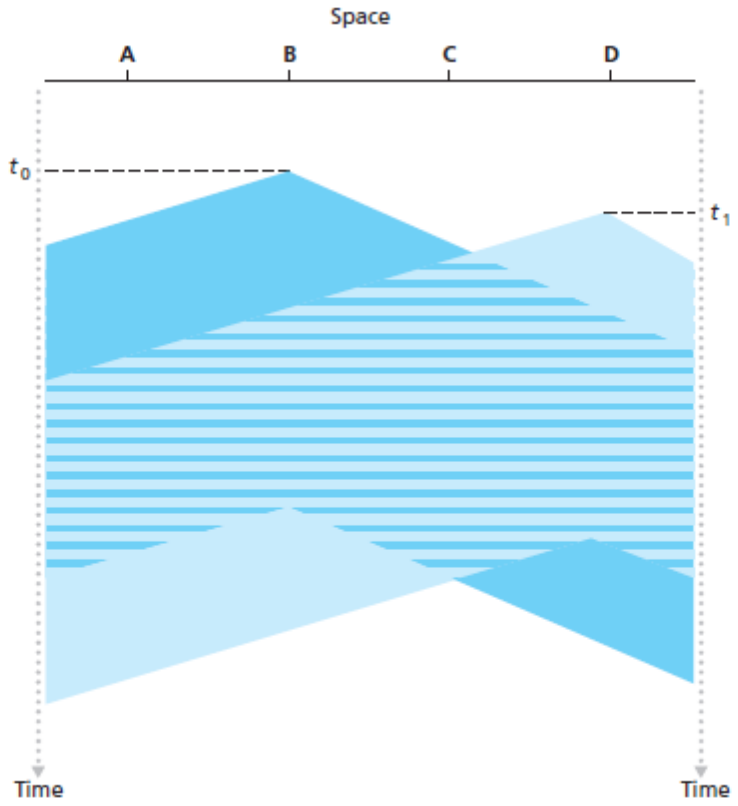


Figure 6.12 ♦ Space-time diagram of two CSMA nodes with colliding transmissions

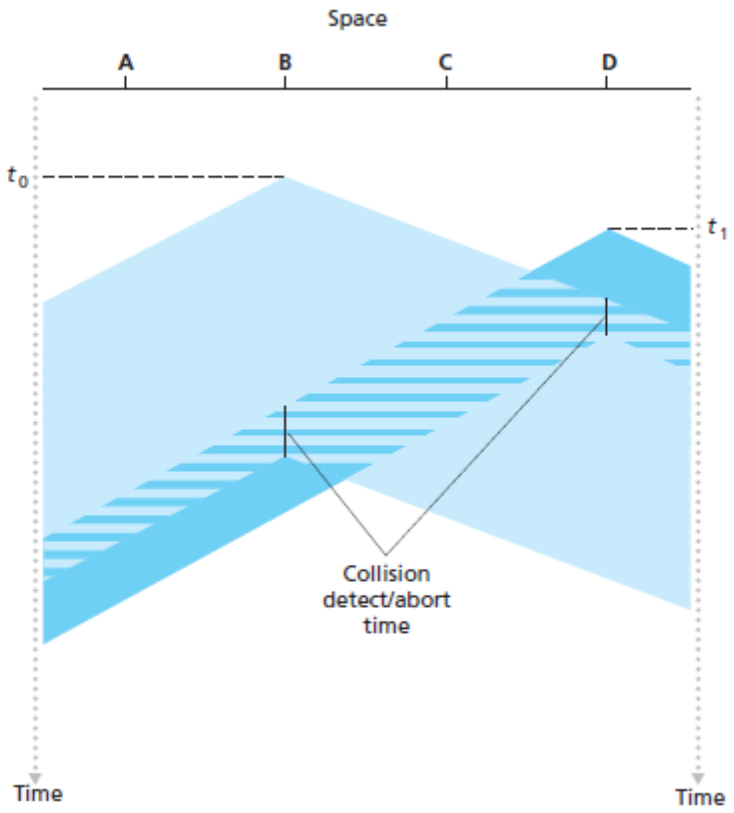


Figure 6.13 ♦ CSMA with collision detection

CSMA-CD Algorithm:

1. The adapter obtains a datagram from the network layer, prepares a link-layer frame, and puts the frame adapter buffer.
2. If the adapter senses that the channel is idle (that is, there is no signal energy entering the adapter from the channel), it starts to transmit the frame. If, on the other hand, the adapter senses that the channel is busy, it waits until it senses no signal energy and then starts to transmit the frame.
3. While transmitting, the adapter monitors for the presence of signal energy coming from other adapters using the broadcast channel.
4. If the adapter transmits the entire frame without detecting signal energy from other adapters, the adapter is finished with the frame. If, on the other hand, the adapter detects signal energy from other adapters while transmitting, it aborts the transmission (that is, it stops transmitting its frame).
5. After aborting, the adapter waits a random amount of time and then returns to step 2.

❑ Exercise 3:

Two hosts, **A** and **B**, are connected to an Ethernet Network along with some other hosts. **Host A** wishes to send **frame *a*** and **host B** wants to send **frame *b*** to some other hosts on the Ethernet. Suppose that host A has had **5 collisions** while trying to send frame *a* (with some hosts other than host B) and host B has had **3 collisions** while trying to send frame b (with some hosts other than host A). **At time T**, only hosts A and B are the ones that are ready to transmit some data. **All other hosts remain idle**.

a. What is the **maximum number K** that A & B can get to calculate waiting time for retransmission?

b. If **bit time** is 1 μs, **K= 10** for A, **K = 5** for B, calculate **waiting times** for both A and B.

6.4 Switched Local Area Networks

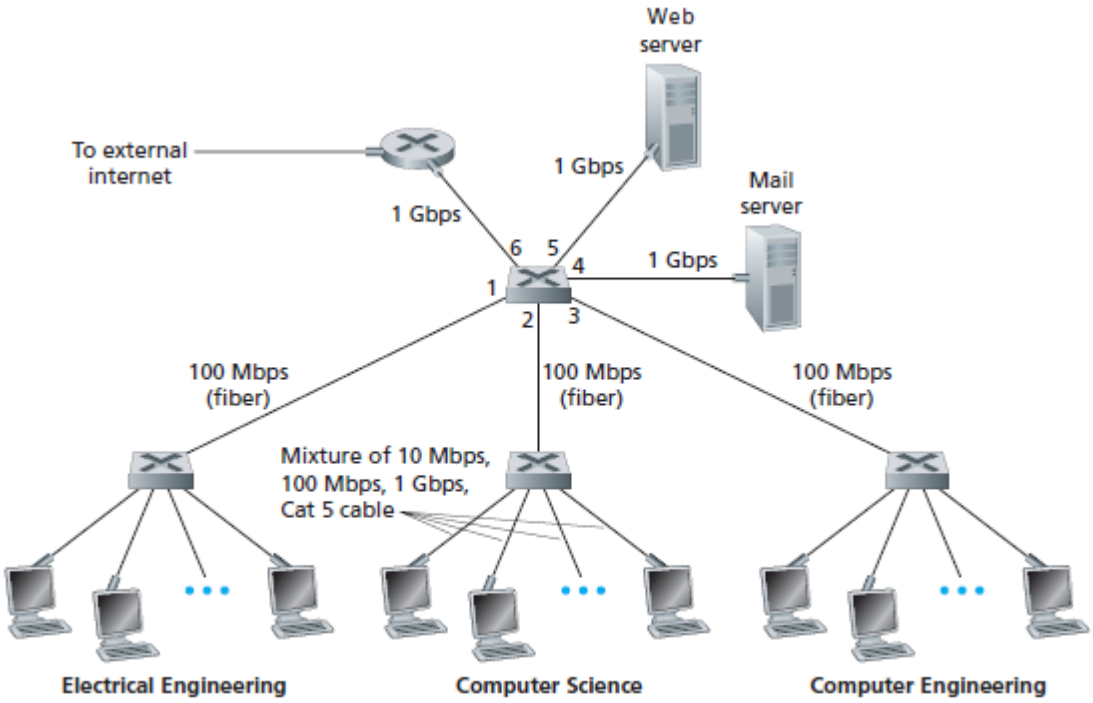


Figure 6.15 ♦ An institutional network connected together by four switches

6.4.1 Link-Layer Addressing and ARP

- **AC Addresses & Address Resolution Protocol (ARP)**

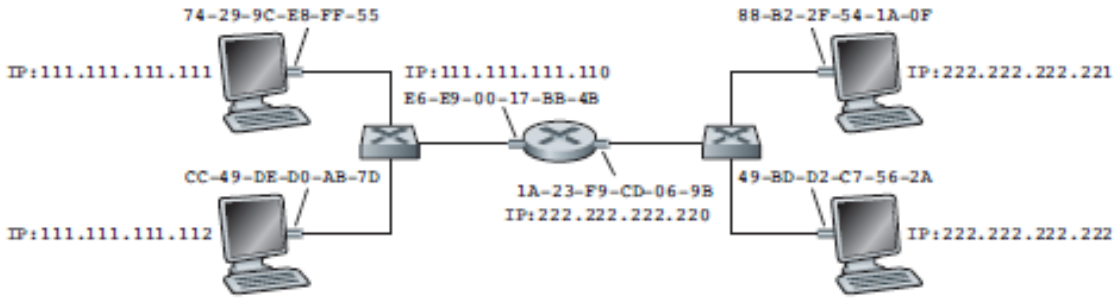
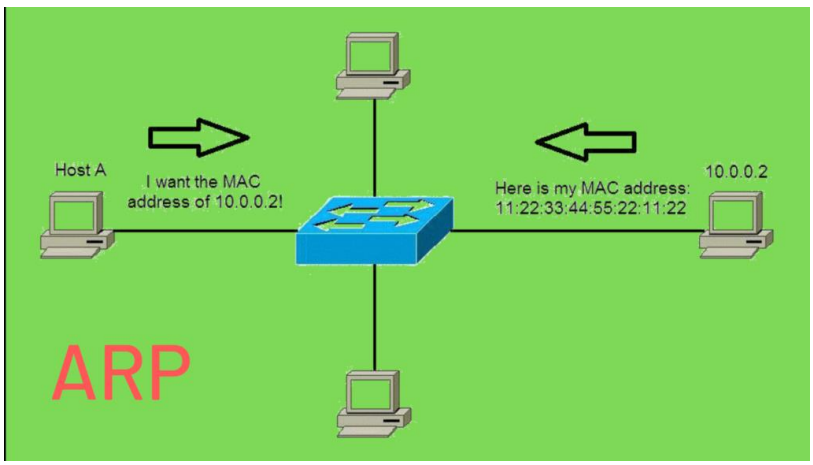
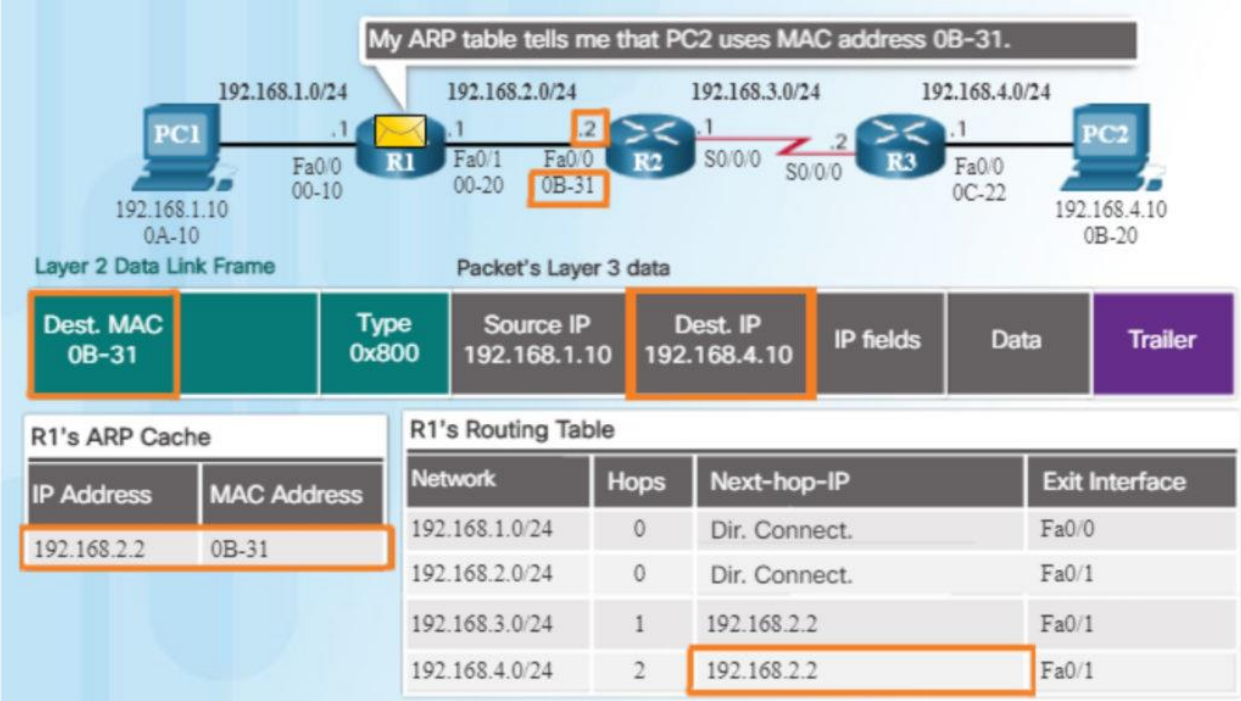


Figure 5.19 ♦ Two subnets interconnected by a router

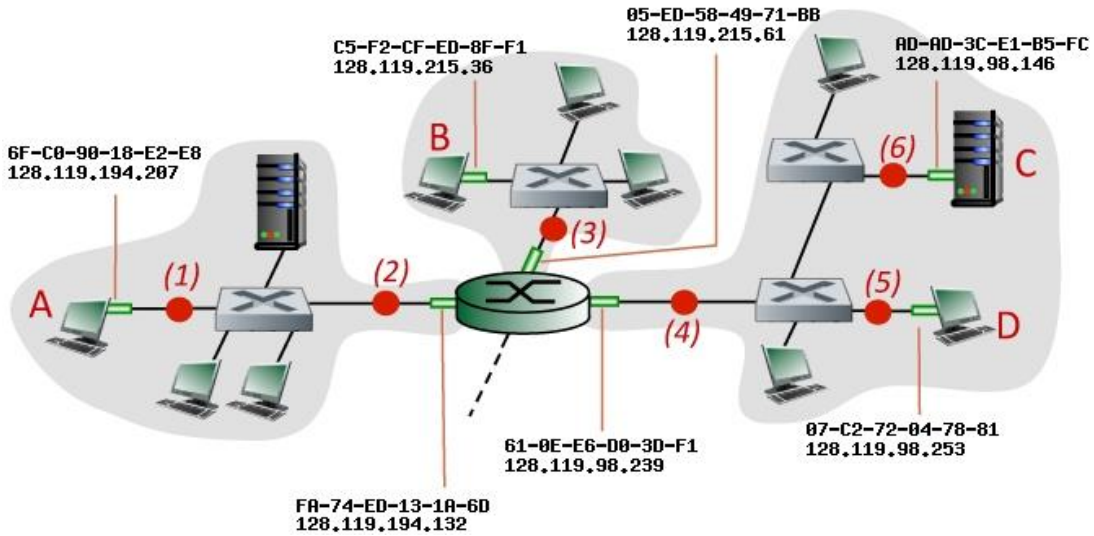




- **Switch Table (MAC Table)**

❑ Exercise 1:

Consider the figure below. The IP and MAC addresses are shown for nodes A, B, C and D, as well as for the router's interfaces. Consider an IP datagram being sent from node **A** to node **C**.



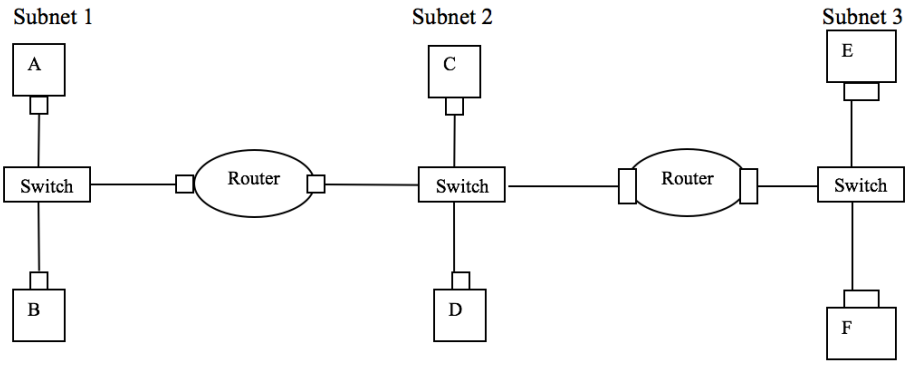
i. Why **ARP** protocol is needed?

ii. Consider an **ARP request** send by **node C**. **Which IP address** will respond to the request?

iii. Now, the **ARP table is filled up**. For the **data packet from C-to-A**, what are the **MAC source** and **destination address** and **IP source and destination addresses** at points **(6)**, **(4)**, **(2)**, and **(1)**?

❑ Exercise 2:

Consider **three LANs** interconnected by two routers, as shown in the following Figure.



a) **Assign IP addresses** to all of the interfaces. For **Subnet 1** use addresses of the form **192.168.1.***, for **Subnet 2** use addresses of the form **192.168.2.***, and for **Subnet 3** use addresses of the form **192.168.3.***.
b) **Assign MAC addresses** to all of the adapters.
c) A host can tell whether another host is on the same LAN by comparing its IP address with that of the other host. Consider **sending an IP packet** from **Host E** to **Host B**. Suppose all of the ARP tables are up to date. **Enumerate all of the steps**.
d) Suppose **the router** between Subnets 1 and 2 is replaced with a switch S1, and label the router between Subnets 2 and 3 as R1. Also suppose that now the ARP tables are not up to date. Will E perform an **ARP query** to find **B's MAC address**? **Why**? In the Ethernet frame (containing the IP packet destined to B), that is delivered to router R1, what are the **source and destination MAC addresses**?

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Knowledge Test # 1

Q.1 Describe **6 (six)** data link layer services.

Q.2 Why **Multiple Access Protocols (MAC)** are used? Describe **3 (three) basic categories of MAC protocols** (Channel partitioning, random access and taking turns).

Q.3 Why **physical (MAC) address** is needed in hardware (LAN card/NIC)?

Q.4 Why **Address Resolution Protocol (ARP)** is used in LANs?

Q.5 What is **Ethernet**?

Q.6 Why **CSMA/CD** is used in LANs?

Q.7 Consider **Figure 5.33**. Now we replace the router between subnets 2 and 3 with a **switch S2**, and label the router between subnets 1 and 2 as **R1**.

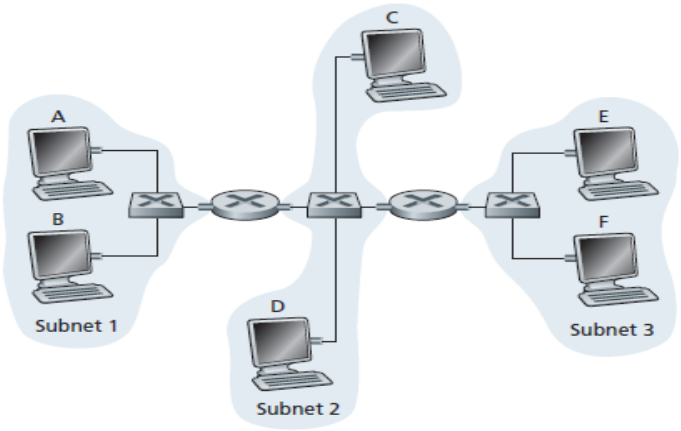


Figure 5.33 ♦ Three subnets, interconnected by routers

a. Consider sending an IP datagram from **Host E** to **Host F**. Will Host E ask router R1 to help forward the datagram? **Why**? In the **Ethernet frame** containing the IP datagram, what are the **source and destination IP and MAC addresses**?

b. Suppose **Host E** would like to send an IP datagram to **Host B**, and assume that **E's ARP cache does not contain B's MAC address**. Will E perform an **ARP query** to find **B's MAC address**? **Why**? In the Ethernet frame (containing the IP datagram destined to B) that is delivered to router R1, what are the **source and destination IP and MAC addresses**?

c. Suppose **Host A** would like to send an IP datagram to **Host B**, and neither A's ARP cache contains B's MAC address nor does B's ARP cache contain A's MAC address. Further suppose that the **switch S1's** (switch connecting A and B) forwarding table contains **entries for Host B and router R1** only. Thus, A will broadcast an **ARP request** message.

i. **What actions** will switch S1 perform once it receives the ARP request message?

ii. Will router R1 also receive this ARP request message? If so, will R1 forward the message to Subnet 2?

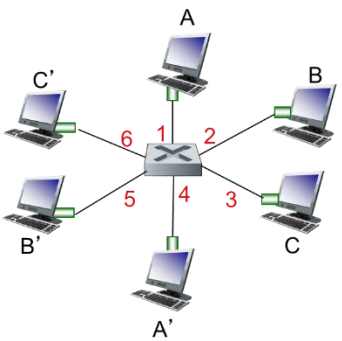
iii. Once Host B receives this ARP request message, it will send back to Host A an ARP response message. But will it send an **ARP query** message to ask for **A's MAC address**? **Why**?

iv. What will **switch S1** do once it receives an **ARP response** message from **Host B**?

Q.8 Suppose nodes *A* and *B* are on the same **10 Mbps broadcast channel**, and the **propagation delay** between the two nodes is **325 bit times**. Suppose CSMA/CD and Ethernet packets are used for this broadcast channel. Suppose **node *A*** begins transmitting a frame and, before it finishes, node *B* begins transmitting a frame. **Can *A* finish transmitting** before it detects that *B* has transmitted? **Why or why not?** If the answer is yes, then *A* incorrectly believes that its frame was successfully transmitted without a collision. [***Hint:*** Suppose at time *t* = 0 bits, *A* begins transmitting a frame. In the worst case, *A* transmits a minimum-sized frame of 512+64 bit times. So *A* would finish transmitting the frame at *t* = 512+64 bit times. Thus, the answer is no, if *B?s* signal reaches *A* before bit time *t* = 512 + 64 bits. In the worst case, when does *B?s* signal reach *A*?]

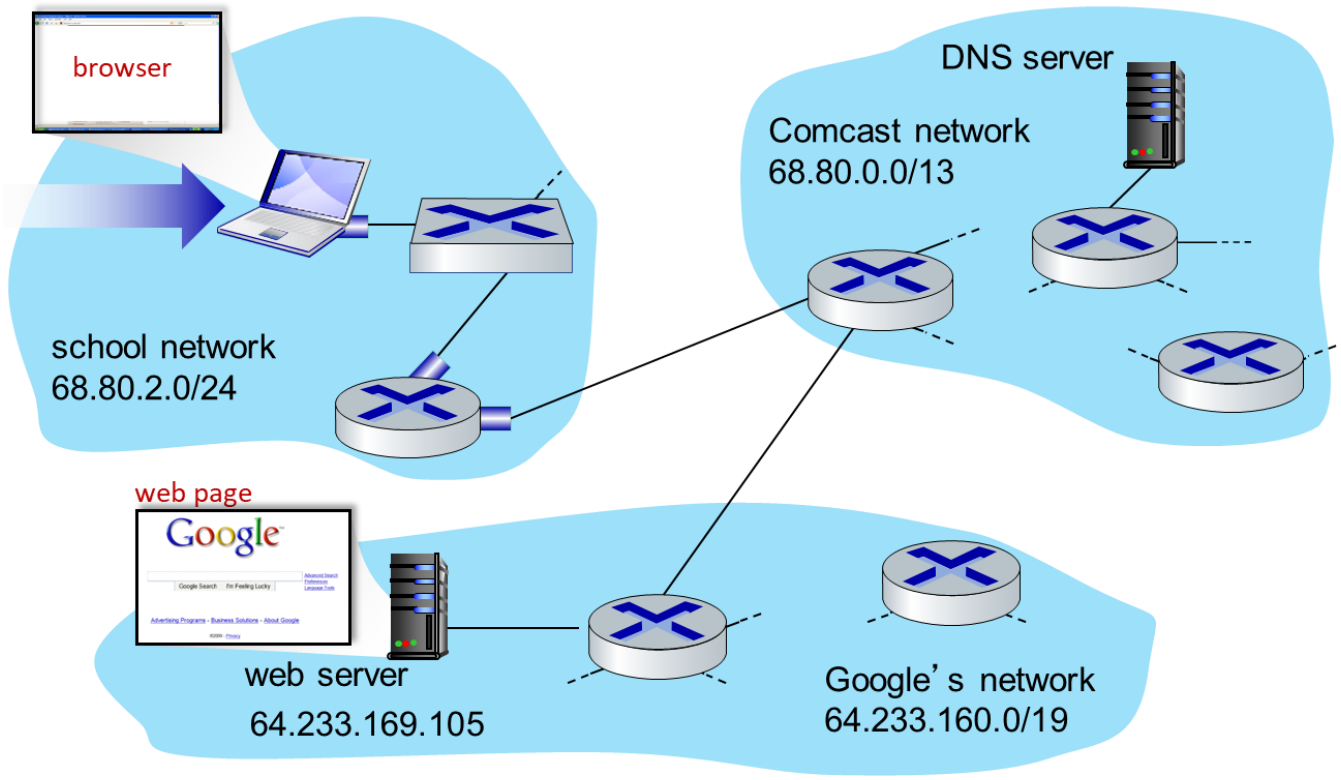
Q.9 Suppose nodes *A* and *B* are on the same **10 Mbps broadcast channel**, and the propagation delay between the two nodes is **245 bit times**. Suppose *A* and *B* send Ethernet frames at the same time, the frames collide, and then *A* and *B* choose different values of *K* in the CSMA/CD algorithm. Assuming no other nodes are active, **can the retransmissions from *A* and *B* collide**? For our purposes, it suffices to work out the following example. Suppose *A* and *B* begin transmission at *t* = 0 bit times. They both detect collisions at *t* = 245 bit times. Suppose $K_A = 0$ and $K_B = 1$. At what time does *B* schedule its retransmission? At what time does *A* begin transmission? At what time does *A's* signal reach *B*? Does *B* refrain from transmitting at its scheduled time?

Q.10 Consider following diagram (the picture with 6 nodes connected in a star topology to a central bridge). Suppose that (i) A sends a frame to D, (ii) D replies with a frame to A, (iii) C sends a frame to D, (iv) D replies with a frame to C. The switch table is initially empty. Show the state of the switch table before and after each of these events. For each of these events, identify the link(s) on which the transmitted frame will be forwarded, and briefly justify your answers.



Action	Switch table state	Link(s) packet is forwarded to	Explanation
A sends a frame to D			
D replies with a frame to A			
C sends a frame to D			
D replies with a frame to C			
A sends a frame to B'			
C' sends a frame to C			

6.7 Retrospective: A Day in the Life of a Web Page Request



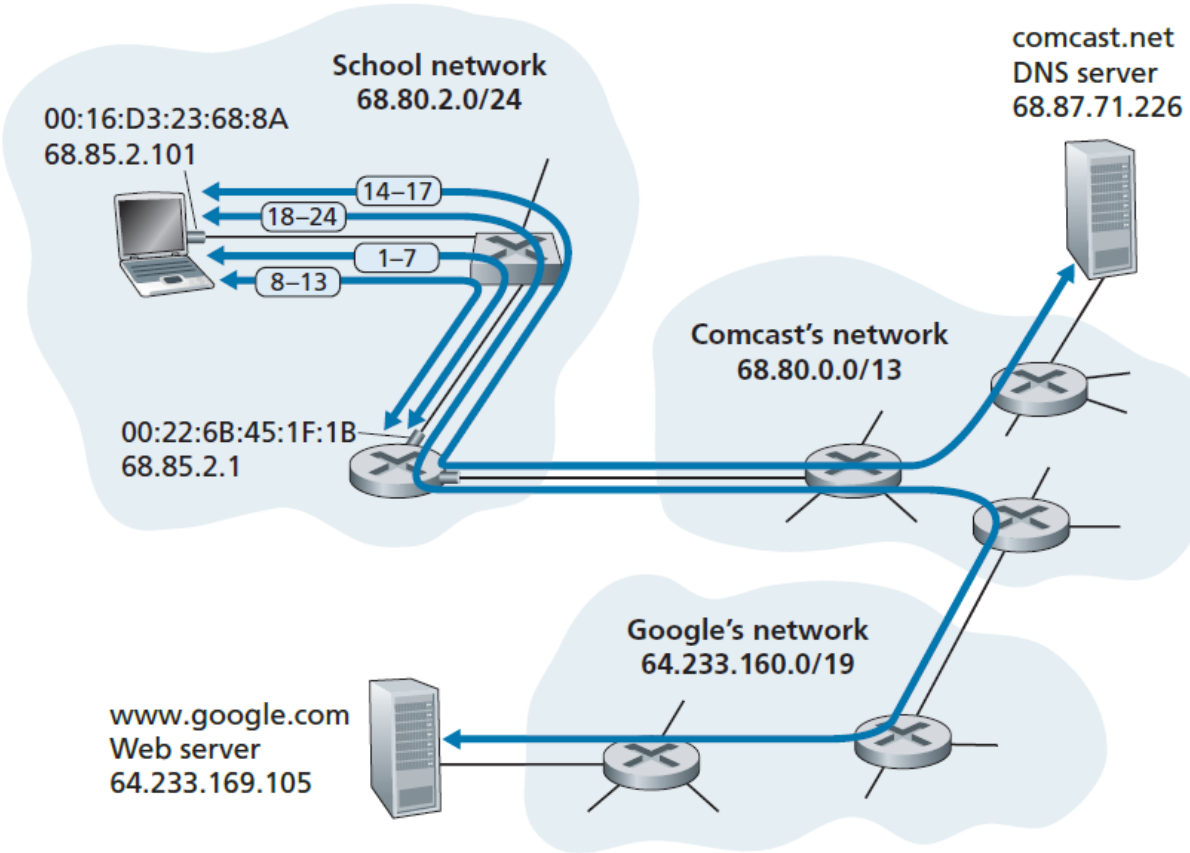


Figure 6.32 ♦ A day in the life of a Web page request: Network setting and actions